

Web Cache Overflow: Exploiting Imprecise Keys for Cache Degradation and Beyond (Code)

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Content

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This reference is a source-code repository. The archive preserves its documentation at an exact commit; the code itself stays in a private mirror and is never checked out, built or run.

- Repository: <<https://github.com/Golim/web-cache-overflow>>
- Commit: 5b336dcdad3c753f84c4c7ca8a6cc667cd6ff0dd
- Documents preserved: 2

LICENSE

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README.md

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Web Cache Overflow: Exploiting Imprecise Keys for Cache Degradation and Beyond

This repository contains the code for the paper "*Web Cache Overflow: Exploiting Imprecise Keys for Cache Degradation and Beyond*", available as a preprint on arXiv (URL coming soon).

Repository Contents

The main directory contains the implementation of the cache overflow attack and several subdirectories, each corresponding to a specific aspect of the research.

cache-overflow.py

Implementation of the cache overflow attack.

penetration-testing/

This directory contains the code for the penetration testing tool designed to identify imprecise cache keys in web applications. It includes scripts for testing various cache-busting techniques, a sample YAML configuration file for defining custom cache-busters, and documentation on how to use the tool. More details can be found in the `penetration-testing/README.md` file.

cache-deduplication/

This directory includes the code for the experiment to estimate the cost of hash-based deduplication. More details can be found in the `cache/deduplication/README.md` file.

cache-poisoning/

This directory contains the code and results for the experiment on cache poisoning facilitated by Web Cache Overflow (WCO). It includes Python scripts, Dockerfiles for setting up the environment, and output logs under the `output/` subdirectory. More details can be found in the `cache/poisoning/README.md` file.

eviction-algorithms/

This directory contains the code and results of the experiment about different cache eviction policies using the Squid proxy. More details can be found in the `eviction-algorithms/README.md`

file.

github-search/

This directory includes scripts and data used to search and analyze open-source cache configurations, along with per-proxy result files. More details can be found in the `github-search/README.md` file.

object-sizes/

This directory contains scripts for WCO experiments with different object sizes. More details can be found in the `object-sizes/README.md` file.

wco-common-defaults/

This directory contains the code for the experiment with different proxies with their common default configurations. It also provides details about Web Polygraph, the tool used to generate the workloads for this experiment, and how to use it. More details can be found in the `wco-common-defaults/README.md` file.

wco-varying-parameters/

This directory contains the scripts and configuration files to run the parameterized experiments for varying cache and object sizes. More details can be found in the `wco-varying-parameters/README.md` file.

utils/

This directory contains analysis helpers for bandwidth, cache poisoning, and console output from Web Polygraph. The scripts come with a `--help` argument to explain their usage.